

Adaptive Multi-Source Information Retrieval for Explainable Soil Health Intelligence Using IoT Sensor Networks and Scientific Knowledge

Angel Oza

B.Tech (ICT), Marwadi University — Batch 2027

1. Problem Statement

Soil health is a critical determinant of agricultural productivity, environmental sustainability, and food security. Conventional soil health monitoring approaches rely predominantly on either periodic laboratory testing or standalone IoT sensor deployments that capture parameters such as soil moisture, temperature, pH, and nutrient concentration. While IoT-based sensing enables continuous, real-time data acquisition, raw sensor readings alone are insufficient to produce reliable, context-aware soil health assessments, since they do not account for crop-specific requirements, soil-type variability, seasonal dynamics, or the established agronomic knowledge documented in scientific literature.

Existing decision-support systems for precision agriculture typically operate on a single data source, offer little or no integration with external scientific knowledge repositories, and function as opaque, black-box models that issue recommendations without a transparent rationale — reducing farmer trust and adoption. There is, therefore, a clear need for an adaptive multi-source information retrieval framework that can fuse real-time IoT sensor streams with relevant scientific and agronomic knowledge, dynamically prioritize the most reliable and contextually relevant sources, and generate explainable, actionable soil health recommendations. This research addresses this gap by proposing such a framework.

2. Objectives

- Design an IoT-based soil health monitoring framework.
- Integrate real-time IoT sensor data with scientific agricultural knowledge.
- Develop an adaptive multi-source information retrieval mechanism.
- Generate explainable soil health recommendations.
- Evaluate the proposed framework using publicly available datasets.

3. Literature Review

3.1 Introduction

Paper	Domain	Methodology	Key Results	Relevance to This Research
Wang, Cai & Zhou (2025) — "A Big Data-Driven Multimodal Information Fusion and Adaptive Reasoning Method for Intelligent Power Grid Inspection,"	Power grid inspection (industrial IoT)	Cross-modal semantic alignment of text, image and tabular data; adaptive dynamic routing (mixture-of-experts) to weight modalities by task context; evidence-provenance module for explainability	86% text-image similarity, 89% retrieval MAP, 81% evidence-consistency accuracy; outperforms a CNN+LSTM late-fusion baseline across all metrics	Demonstrates a working adaptive multi-source fusion architecture (dynamic routing + evidence provenance) that is directly transferable to fusing IoT sensor streams with scientific/agronomic knowledge for explainable soil health recommendations

Paper	Domain	Methodology	Key Results	Relevance to This Research
IEEE BDDM 2025				
Kang & Ye (2026) — "Retrieval of Aerosol Optical Depth Using Multi-Source Data," IEEE CNML 2026	Environmental / atmospheric remote sensing	ResNet-Transformer Feature Fusion (RTFF): CNN branch for local features, Transformer encoder for global dependencies, combined via a gated (sigmoid) fusion mechanism across multiple sensor-derived data components	R ² improves from 0.68 (single source) to 0.90 (adding polarization) to 0.97 (adding wavelength info); RTFF outperforms ResNet, Transformer and VGG16 baselines	Empirically confirms that progressively fusing heterogeneous sensor-derived sources through gated/adaptive fusion yields large, consistent accuracy gains — supporting the core hypothesis that combining multiple IoT modalities with external knowledge will improve soil health assessment accuracy

3.2 Comparison Table (Current Progress)

3.3 Summary

Although drawn from different application domains — industrial inspection and atmospheric remote sensing — both reviewed papers converge on a common insight: adaptively fusing heterogeneous, multi-source data (via dynamic routing or gated mechanisms) consistently and substantially improves prediction accuracy over single-source or naively-combined baselines. The power grid paper additionally demonstrates that an evidence-provenance layer can make such fusion pipelines explainable and auditable, which is directly relevant to the explainability goal of this research.

Neither paper, however, integrates real-time sensor data with unstructured scientific/textual knowledge, and neither targets soil health or an end-user-facing explainable recommendation output. This reinforces the preliminary research gaps identified below and motivates the proposed framework.

4. Research Gaps

- Existing systems mainly rely on IoT sensor data.
- Limited integration of scientific literature.
- Lack of explainable recommendations.
- Limited multi-source information fusion.
- Static decision-making models.
- Limited validation on diverse datasets.

5. Proposed Methodology

- IoT Sensor Data Collection
- Data Preprocessing
- Multi-Source Information Retrieval

- Information Fusion
- Explainable AI Module
- Soil Health Intelligence
- Recommendation Generation

6. Tentative Architecture

The proposed system is organized as a seven-stage pipeline, moving from raw IoT sensing through adaptive multi-source retrieval and fusion to an explainable, farmer-facing recommendation:

Stage	Module	Function
1	IoT Sensing Layer	Field-deployed sensors (soil moisture, temperature, pH, NPK, EC) stream time-stamped readings
2	Data Preprocessing	Cleaning, denoising, normalization and missing-value handling of raw sensor streams
3	Multi-Source Retrieval Engine	Adaptively retrieves relevant scientific literature, agronomic guidelines and crop-specific knowledge conditioned on current sensor context
4	Information Fusion Module	Gated / attention-based fusion combining sensor embeddings with retrieved knowledge embeddings into a unified representation
5	Explainable AI (XAI) Module	Generates human-interpretable rationales (e.g., feature attribution, evidence traceback) for each assessment
6	Soil Health Intelligence Core	Produces the overall soil health index / status from the fused, explainable representation
7	Recommendation Generator	Translates soil health intelligence into actionable, farmer-facing recommendations

7. Future Work

- Complete the review of 20–25 research papers.
- Prepare the comparative literature review table.
- Refine the research gaps.
- Finalize the proposed architecture.
- Identify suitable datasets for experimentation.

8. References

[1] W. Lin, D. Cai, and J. Zhou, "A big data–driven multimodal information fusion and adaptive reasoning method for intelligent power grid inspection," in Proc. 2025 Int. Conf. Big Data and Data Mining (BDDM), 2025, doi: 10.1109/BDDM68348.2025.11442148.

[2] S. Kang and K. Ye, "Retrieval of aerosol optical depth using multi-source data," in Proc. 2026 Int. Conf. Communication Networks and Machine Learning (CNML), 2026, pp. 676–679, doi: 10.1109/CNML68938.2026.11452285.